

Andreev transport through a ferromagnet–bilayer-graphene–superconductor junction

E. C. Siqueira*

Departamento de Física, Universidade Tecnológica Federal do Paraná (UTFPR), 84016-210, Ponta Grossa, PR, Brasil

A. I. González and P. A. Orellana

Departamento de Física, Universidad Técnica Federico Santa María, 110 V, Valparaíso, Casilla, Chile

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We study Andreev transport through a hybrid graphene junction composed of a ferromagnetic monolayer lead, a finite normal AB-stacked bilayer-graphene region, and an s -wave superconducting monolayer lead. Two atomically distinct contact geometries are considered, depending on whether the superconducting lead continues from the same graphene layer as the ferromagnetic lead or from the opposite layer. Starting from a microscopic tight-binding Hamiltonian, we formulate a Bogoliubov–de Gennes transfer-matrix approach in which the finite bilayer is propagated through flux-normalized modes and the two contacts are treated by local atomic matching conditions. The resulting conductance can be written in terms of a four-dimensional round-trip matrix whose eigenvalues diagnose the finite-length resonance structure responsible for the bright ridges observed in conductance maps. This manuscript draft emphasizes the compact model and formalism; numerical results and ridge overlays will be inserted in the next stage.

I. INTRODUCTION

Andreev reflection converts an incident electron into a reflected hole while transferring a Cooper pair to a superconducting condensate. In graphene-based systems this process is sensitive not only to the superconducting gap, but also to the band structure, transverse momentum, doping, and interface geometry. Adding a ferromagnetic region introduces spin-dependent longitudinal momenta and therefore suppresses or reshapes Andreev reflection through the mismatch between the electron and the time-reversed hole channel.

The system considered here combines these ingredients with a finite AB-stacked bilayer-graphene section. The bilayer acts as an internal coherent cavity: an electron-like or hole-like excitation injected from the ferromagnetic lead can undergo repeated normal and Andreev reflections at the two atomic contacts before escaping. The finite length N then enters through propagation phases and evanescent factors, producing resonance ridges in maps of the conductance as a function of excitation energy and bilayer length.

The aim of this paper is to formulate a microscopic transfer-matrix description of this problem and to use it to interpret the length-dependent Andreev structures. Unlike a phenomenological BTK model with an adjustable barrier strength [2], the contact transparency is determined here by the actual honeycomb connectivity, the layer termination, the exchange field, the chemical potentials, and the superconducting pairing. The normal bilayer transfer construction follows the lattice convention of Ref. 1, generalized here to a spin-resolved Bogoliubov–de Gennes problem.

The central formal result is a round-trip representation of the scattering problem. For each transverse channel K ,

spin sector σ , and geometry g , repeated internal cycles are summed by the resolvent

$$\left[\mathbb{I}_4 - \mathbf{M}_{\text{rt}}^{(g)}(E, K, \sigma, N) \right]^{-1}, \quad (1)$$

where $\mathbf{M}_{\text{rt}}^{(g)}$ is the microscopic round-trip matrix of the finite bilayer cavity. Its eigenvalues provide a direct diagnostic of the bright ridges in the conductance maps: a fixed- K resonance occurs when a bright round-trip eigenvalue approaches unity, while a ridge in the transverse-integrated conductance survives only when the corresponding resonance branch is sufficiently flat over the open incident-channel window.

The manuscript is organized as follows. Section II defines the lattice geometry, Hamiltonian, and BdG sector convention. Section IV constructs the transfer matrices, flux normalization, and stable finite-length propagation. Section V derives the interface maps, round-trip matrix, scattering amplitudes, and conductance. Section VI explains the resonance diagnostics used to interpret the conductance ridges. Section VII is reserved for the numerical maps and overlays.

II. MODEL

A. Geometry and physical domains

The system is illustrated in Fig. 1. It is composed of two graphene layers, labeled by $j = 1, 2$, arranged in two contact geometries denoted by $1 \rightarrow 1$ and $1 \rightarrow 2$, corresponding to Figs. 1(a) and 1(b), respectively. The longitudinal and transverse directions are indexed by the integers m and n . With this convention, the finite bilayer region occupies

$$1 \leq m \leq N \quad (2)$$

* ecosta@utfpr.edu.br

in both geometries. For $m \leq 0$ the graphene monolayer is ferromagnetic, whereas for $m \geq N + 1$ the monolayer is an s -wave superconductor.

The main difference between the two geometries concerns the right contact, where the finite bilayer is coupled to the superconducting monolayer. In the $1 \rightarrow 1$ geometry, the superconductor is coupled to the lower layer, the same layer that is connected to the ferromagnetic monolayer. In the $1 \rightarrow 2$ geometry, the ferromagnetic monolayer is still connected to the lower layer, but the superconducting monolayer is coupled to the upper layer. This distinction is most clearly visualized in the block diagram of Fig. 1(d). In both geometries we assume Bernal stacking in the central region, as shown in Fig. 1(c), with an A atom of the lower layer coupled to a B atom of the upper layer by the interlayer hopping $t_\perp$.

We now translate this figure-based description into the domains used in the Hamiltonian sums. The set of layer-cell pairs that physically exist in the $1 \rightarrow 1$ geometry is

$$\mathcal{D}_{1 \rightarrow 1} = \{(1, m) : m \in \mathbb{Z}\} \cup \{(2, m) : 1 \leq m \leq N\}. \quad (3)$$

Thus layer 1 extends through the whole junction, while layer 2 exists only in the finite bilayer overlap. For the $1 \rightarrow 2$ geometry,

$$\mathcal{D}_{1 \rightarrow 2} = \{(1, m) : m \leq N\} \cup \{(2, m) : m \geq 1\}. \quad (4)$$

Thus layer 1 extends from the ferromagnetic lead through the central bilayer and terminates at the right edge, while layer 2 begins at the left edge of the central bilayer and continues into the superconducting lead. In what follows, $g \in \{1 \rightarrow 1, 1 \rightarrow 2\}$ labels the chosen geometry and $\mathcal{D}_g$ denotes the corresponding physical domain.

All microscopic sums below are restricted to $(j, m) \in \mathcal{D}_g$. A hopping or pairing term is included only if every site appearing in that term belongs to the physical domain. A missing neighbor at a terminated layer is therefore not assigned a zero amplitude by hand; the corresponding bond is absent from the Hamiltonian. This convention is the microscopic origin of the termination conditions used later in the interface matching.

The remaining internal labels are the sublattice index $\alpha = A, B$ and the spin label $\sigma = \uparrow, \downarrow$, with

$$s_\uparrow = +1, \quad s_\downarrow = -1. \quad (5)$$

B. Real-space tight-binding Hamiltonian

The microscopic Hamiltonian is written as

$$\hat{H}^{(g)} = \hat{H}_t^{(g)} + \hat{H}_\perp + \hat{H}_{\text{on}}^{(g)} + \hat{H}_\Delta^{(g)}. \quad (6)$$

The superscript (g) emphasizes that the intralayer, onsite, and pairing terms depend on the physical domain of the chosen contact geometry.

With the honeycomb convention used in Ref. 1, each A site is connected to three nearest-neighbor B sites. The intralayer hopping is

$$\begin{aligned} \hat{H}_t^{(g)} = -t \sum_{(j,m) \in \mathcal{D}_g} \sum_{n,\sigma} & \left[\hat{a}_{jmn\sigma}^\dagger \hat{b}_{jmn\sigma} + \hat{a}_{jmn\sigma}^\dagger \hat{b}_{j,m+1,n,\sigma} \right. \\ & \left. + \hat{a}_{jmn\sigma}^\dagger \hat{b}_{j,m,n+1,\sigma} + \text{H.c.} \right]. \end{aligned} \quad (7)$$

The longitudinal bond in Eq. (7) is present only when both (j, m) and $(j, m + 1)$ belong to $\mathcal{D}_g$.

The AB interlayer hopping exists only inside the central bilayer, where both layers are present:

$$\hat{H}_\perp = -t_\perp \sum_{m=1}^N \sum_{n,\sigma} \left(\hat{a}_{1mn\sigma}^\dagger \hat{b}_{2mn\sigma} + \text{H.c.} \right). \quad (8)$$

Thus the dimer bond connects A_1 to B_2 for $1 \leq m \leq N$.

The onsite normal term is

$$\hat{H}_{\text{on}}^{(g)} = \sum_{(j,m) \in \mathcal{D}_g} \sum_{n,\alpha,\sigma} [V_j(m) - \mu(m) - s_\sigma h(m)] \hat{n}_{jmn\alpha\sigma}, \quad (9)$$

where

$$\hat{n}_{jmn\alpha\sigma} = \hat{c}_{jmn\alpha\sigma}^\dagger \hat{c}_{jmn\alpha\sigma}. \quad (10)$$

Here $\alpha = A, B$ is the sublattice index. The spatial region is determined by m and by the physical domain $\mathcal{D}_g$, not by an additional operator label. The chemical potential is piecewise constant,

$$\mu(m) = \begin{cases} \mu_F, & m \leq 0, \\ \mu_C, & 1 \leq m \leq N, \\ \mu_S, & m \geq N + 1, \end{cases} \quad (11)$$

and the collinear exchange field is confined to the ferromagnetic lead,

$$h(m) = \begin{cases} h, & m \leq 0, \\ 0, & m \geq 1. \end{cases} \quad (12)$$

The layer potentials $V_1(m)$ and $V_2(m)$ are nonzero only in the central bilayer and allow for layer-resolved gating.

The superconducting lead contains onsite spin-singlet pairing,

$$\begin{aligned} \hat{H}_\Delta^{(g)} = \sum_{(j,m) \in \mathcal{D}_g} \sum_{n,\alpha} & \left[\Delta(m) \hat{c}_{jmn\alpha\uparrow}^\dagger \hat{c}_{jmn\alpha\downarrow}^\dagger \right. \\ & \left. + \Delta^*(m) \hat{c}_{jmn\alpha\downarrow} \hat{c}_{jmn\alpha\uparrow} \right]. \end{aligned} \quad (13)$$

where

$$\Delta(m) = \begin{cases} 0, & m \leq N, \\ \Delta_0 e^{i\varphi}, & m \geq N + 1. \end{cases} \quad (14)$$

Because the sum in Eq. (13) is restricted to $\mathcal{D}_g$, the pair potential is placed only on the layer that physically continues into the superconducting lead.

C. Fourier-space Hamiltonian

We now use translational invariance along the transverse direction to write the model in momentum space. For N_y transverse cells with periodic boundary conditions, define the dimensionless transverse momentum $K = k_y a_y \in [-\pi, \pi]$ and use

$$\begin{aligned}\hat{a}_{jmn\sigma} &= \frac{1}{\sqrt{N_y}} \sum_K e^{iKn} \hat{a}_{jmK\sigma}, \\ \hat{b}_{jmn\sigma} &= \frac{1}{\sqrt{N_y}} \sum_K e^{iKn} \hat{b}_{jmK\sigma}.\end{aligned}\quad (15)$$

The two intralayer bonds that do not change m combine into the effective intracell hopping

$$\eta(K) = t(1 + e^{iK}), \quad \xi = t, \quad (16)$$

where ξ denotes the longitudinal hopping across one transverse step. The intralayer term becomes

$$\begin{aligned}\hat{H}_t^{(g)}(K) &= - \sum_{(j,m) \in \mathcal{D}_g} \sum_{\sigma} \left[\eta(K) \hat{a}_{jmK\sigma}^\dagger \hat{b}_{jmK\sigma} \right. \\ &\quad \left. + \xi \hat{a}_{jmK\sigma}^\dagger \hat{b}_{j,m+1,K,\sigma} + \text{H.c.} \right].\end{aligned}\quad (17)$$

As in real space, the longitudinal bond is included only when the two sites connected by it exist in $\mathcal{D}_g$.

The interlayer hopping is diagonal in K because the dimer bond is vertical within the same cell:

$$\hat{H}_\perp(K) = -t_\perp \sum_{m=1}^N \sum_{\sigma} \left(\hat{a}_{1mK\sigma}^\dagger \hat{b}_{2mK\sigma} + \text{H.c.} \right). \quad (18)$$

The onsite normal term is also diagonal in K ,

$$\begin{aligned}\hat{H}_{\text{on}}^{(g)}(K) &= \sum_{(j,m) \in \mathcal{D}_g} \sum_{\alpha,\sigma} [V_j(m) - \mu(m) - s_\sigma h(m)] \\ &\quad \times \hat{n}_{jmK\alpha\sigma}.\end{aligned}\quad (19)$$

The pairing term is different: because it contains two creation or two annihilation operators at the same transverse cell, it couples opposite transverse momenta,

$$\begin{aligned}\hat{H}_\Delta^{(g)}(K) &= \sum_{(j,m) \in \mathcal{D}_g} \sum_{\alpha} \left[\Delta(m) \hat{c}_{jmK\alpha\uparrow}^\dagger \hat{c}_{jm,-K,\alpha\downarrow}^\dagger \right. \\ &\quad \left. + \Delta^*(m) \hat{c}_{jm,-K,\alpha\downarrow} \hat{c}_{jmK\alpha\uparrow} \right].\end{aligned}\quad (20)$$

Thus the normal part is diagonal in K , while the singlet-pairing part closes only after K and $-K$ are treated together. The independent scattering channel labeled by K is therefore the BdG sector

$$(K, \sigma) \oplus (-K, -\sigma), \quad (21)$$

not an ordinary normal-state single-momentum block. In this sense the Hamiltonian is block diagonal in paired BdG sectors; no additional independent sum over $-K$ is introduced when the scattering problem is solved at fixed K .

III. BOGOLIUBOV-DE GENNES EQUATIONS

The Fourier-space Hamiltonian of Sec. II C is now converted into algebraic tight-binding equations. This intermediate step is useful because it shows explicitly how the microscopic model becomes the transfer recurrence used in Sec. IV.

A. BdG sector and quasiparticle operator

For a fixed electron spin σ , the corresponding hole has spin $-\sigma$. We use the quasiparticle operator

$$\hat{\gamma}_{EK\sigma}^\dagger = \sum_{j,m,\alpha} \left[u_{j\alpha}(m) \hat{c}_{jmK\alpha\sigma}^\dagger + s_\sigma v_{j\alpha}(m) \hat{c}_{jm,-K,\alpha,-\sigma} \right]. \quad (22)$$

The factor s_σ in the hole component cancels the spin-dependent sign produced by the spin-singlet pairing commutator. With this convention, the two independent sectors $(e_\uparrow, h_\downarrow)$ and $(e_\downarrow, h_\uparrow)$ have the same off-diagonal BdG entries $+\Delta$ and $+\Delta^*$.

The eigenoperator equation

$$[\hat{H}^{(g)}, \hat{\gamma}_{EK\sigma}^\dagger] = E \hat{\gamma}_{EK\sigma}^\dagger \quad (23)$$

gives, in compact one-particle notation,

$$\begin{pmatrix} H_\sigma(K) & \Delta \\ \Delta^\dagger & -H_{-\sigma}^*(-K) \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = E \begin{pmatrix} u \\ v \end{pmatrix}. \quad (24)$$

The lower block contains three operations: spin reversal, momentum reversal, and complex conjugation. These operations are responsible for the signs in the hole equations below.

B. Monolayer BdG equations

For a uniform monolayer region with parameters (μ, h, Δ) and no layer gate potential, define

$$\epsilon_{e,\sigma} = E + \mu + s_\sigma h, \quad \epsilon_{h,\sigma} = E - \mu + s_\sigma h. \quad (25)$$

Projecting Eq. (23) onto the A and B sublattices gives

$$\epsilon_{e,\sigma} u_A(m) = -\eta u_B(m) - \xi u_B(m+1) + \Delta v_A(m), \quad (26)$$

$$\epsilon_{e,\sigma} u_B(m) = -\eta^* u_A(m) - \xi^* u_A(m-1) + \Delta v_B(m), \quad (27)$$

$$\epsilon_{h,\sigma} v_A(m) = +\eta v_B(m) + \xi v_B(m+1) + \Delta^* u_A(m), \quad (28)$$

$$\epsilon_{h,\sigma} v_B(m) = +\eta^* v_A(m) + \xi^* v_A(m-1) + \Delta^* u_B(m). \quad (29)$$

The electron equations inherit the minus sign of the hopping Hamiltonian, whereas the hole equations have the opposite hopping sign because they come from the block $-H_{-\sigma}^*(-K)$. The exchange field enters both $\epsilon_{e,\sigma}$ and $\epsilon_{h,\sigma}$ with the same sign because the hole is a missing electron of spin $-\sigma$.

In the ferromagnetic lead $\Delta = 0$ and $h \neq 0$. In the superconducting lead $h = 0$ and $\Delta = \Delta_0 e^{i\varphi}$.

C. Central bilayer equations

Inside the central bilayer, $h = \Delta = 0$. Define the layer-dependent scalar parameters

$$x_{e,j} = E + \mu_C - V_j, \quad x_{h,j} = \mu_C - E - V_j. \quad (30)$$

The electron amplitudes satisfy

$$x_{e,1}u_{A1}(m) = -\eta u_{B1}(m) - \xi u_{B1}(m+1) - t_\perp u_{B2}(m), \quad (31)$$

$$x_{e,1}u_{B1}(m) = -\eta^* u_{A1}(m) - \xi^* u_{A1}(m-1), \quad (32)$$

$$x_{e,2}u_{A2}(m) = -\eta u_{B2}(m) - \xi u_{B2}(m+1), \quad (33)$$

$$x_{e,2}u_{B2}(m) = -\eta^* u_{A2}(m) - \xi^* u_{A2}(m-1) - t_\perp u_{A1}(m). \quad (34)$$

The interlayer hopping appears only in the A_1 and B_2 equations, reflecting the AB dimer bond shown in Fig. 1(c).

The central hole equations have the same algebraic structure after the replacement

$$u \mapsto v, \quad x_{e,j} \mapsto x_{h,j}. \quad (35)$$

This is the reason the central BdG transfer matrix in the next section is a direct sum of two normal bilayer transfer matrices evaluated at the two scalar arguments $\mu_C + E$ and $\mu_C - E$.

IV. TRANSFER-MATRIX AND MODE FORMALISM

The lattice equations of Sec. III are second-order recurrences in the longitudinal index: an equation at cell m contains amplitudes at $m-1$, m , and $m+1$. The transfer-matrix construction rewrites them as first-order recurrences by grouping amplitudes on the two sides of a longitudinal cut.

A. Staggered transfer states

For a normal monolayer, the natural transfer state is not $(A(m), B(m))^T$ but the staggered pair

$$\phi_m = \begin{pmatrix} A(m-1) \\ B(m) \end{pmatrix}, \quad \phi_{m+1} = \begin{pmatrix} A(m) \\ B(m+1) \end{pmatrix}. \quad (36)$$

This choice is dictated by Eqs. (26) and (27): the A equation contains $B(m+1)$, while the B equation contains $A(m-1)$.

For the normal bilayer, the corresponding state is

$$\Psi_m = \begin{pmatrix} A_1(m-1) \\ B_1(m) \\ A_2(m-1) \\ B_2(m) \end{pmatrix}, \quad \Psi_{m+1} = \begin{pmatrix} A_1(m) \\ B_1(m+1) \\ A_2(m) \\ B_2(m+1) \end{pmatrix}. \quad (37)$$

The central BdG transfer state is the direct electron-hole extension

$$\Xi_m = \begin{pmatrix} u_{A1}(m-1) \\ u_{B1}(m) \\ u_{A2}(m-1) \\ u_{B2}(m) \\ v_{A1}(m-1) \\ v_{B1}(m) \\ v_{A2}(m-1) \\ v_{B2}(m) \end{pmatrix}. \quad (38)$$

The same staggered ordering is used everywhere below; changing it would also change the signs and the selector components used at the interfaces.

B. From tight-binding equations to transfer matrices

The connection between the algebraic tight-binding equations and the transfer matrix can be seen directly in the central bilayer. Let $x_j = \epsilon - V_j$, where ϵ is the scalar energy argument of the normal bilayer problem. Rearranging Eqs. (31)–(34) with the amplitudes belonging to Ψ_{m+1} on the left and those belonging to Ψ_m on the right gives

$$L_{BL}\Psi_{m+1} = R_{BL}\Psi_m, \quad T_{BL}(\epsilon, K) = L_{BL}^{-1}R_{BL}. \quad (39)$$

The matrices are

$$L_{BL} = \begin{pmatrix} x_1 & \xi & 0 & 0 \\ \eta^* & 0 & 0 & 0 \\ 0 & 0 & x_2 & \xi \\ t_\perp & 0 & \eta^* & 0 \end{pmatrix}, \quad (40)$$

$$R_{BL} = \begin{pmatrix} 0 & -\eta & 0 & -t_\perp \\ -\xi^* & -x_1 & 0 & 0 \\ 0 & 0 & 0 & -\eta \\ 0 & 0 & -\xi^* & -x_2 \end{pmatrix}. \quad (41)$$

The first row of Eq. (40), for example, is just Eq. (31) written as

$$x_1 A_1(m) + \xi B_1(m+1) = -\eta B_1(m) - t_\perp B_2(m). \quad (42)$$

Thus the transfer matrix is not an additional approximation; it is a reorganization of the same tight-binding equations.

The normal bilayer modes may be written in Bloch form. The corresponding transfer roots are

$$\lambda_{\nu,\pm} = \exp \left[i \left(\frac{K}{2} \pm Q_\nu \right) \right], \quad (43)$$

where Q_ν is determined by the bilayer secular equation. In the numerical implementation the eigenvectors are obtained from the transfer matrix itself rather than from closed-form spinors, because special points such as layer charge neutrality may make intermediate analytic divisions singular.

C. Central BdG transfer matrix

Since the central region is normal, electron and hole amplitudes propagate independently. Using Eq. (35), the central BdG transfer matrix is

$$\Xi_{m+1} = T_C(E, K)\Xi_m, \quad (44)$$

$$T_C(E, K) = \text{diag}(T_e, T_h), \quad (45)$$

$$T_e = T_{\text{BL}}(\mu_C + E, K), \quad (46)$$

$$T_h = T_{\text{BL}}(\mu_C - E, K).$$

The explicit determinant checks and limiting cases of T_{BL} are collected in Appendix B. Lead transfer matrices T_F and T_S are constructed from the monolayer BdG equations, Eqs. (26)–(29), by the same staggered procedure.

D. Flux metric and mode classification

The propagation direction of a multiband transfer eigenmode must be determined by its lattice current, not by the sign of a phase label alone. For the monolayer BdG ordering

$$\Phi_m = (u_A(m-1) \ u_B(m) \ v_A(m-1) \ v_B(m))^T, \quad (47)$$

the quasiparticle current can be written as the bilinear form

$$J_{\text{qp}}(\Phi) = \Phi^\dagger \Sigma \Phi, \quad (48)$$

$$\Sigma = \begin{pmatrix} \sigma_\xi & 0 \\ 0 & -\sigma_\xi \end{pmatrix}, \quad \sigma_\xi = \begin{pmatrix} 0 & -i\xi \\ i\xi^* & 0 \end{pmatrix}.$$

For the central ordering of Eq. (38),

$$\Sigma_C = \text{diag}(\sigma_\xi, \sigma_\xi, -\sigma_\xi, -\sigma_\xi). \quad (49)$$

Current conservation in a uniform region implies the pseudo-unitarity relation

$$T^\dagger \Sigma T = \Sigma. \quad (50)$$

If $T\zeta_i = \lambda_i\zeta_i$ and $T\zeta_j = \lambda_j\zeta_j$, Eq. (50) gives

$$(\lambda_i^*\lambda_j - 1)\zeta_i^\dagger \Sigma \zeta_j = 0. \quad (51)$$

Thus distinct propagating modes can be chosen flux-orthogonal. Degenerate propagating subspaces should be explicitly orthogonalized with respect to Σ before squared modal amplitudes are interpreted as probabilities.

In the left ferromagnetic lead, the prescribed incident electron is a unit-flux mode ζ_{in} with $J > 0$. The outgoing lead basis Z_F^- contains reflected propagating modes with $J < 0$ and evanescent modes decaying toward $m \rightarrow -\infty$. At the left boundary,

$$\Phi_F(1) = \zeta_{\text{in}} + Z_F^- r. \quad (52)$$

In the right superconducting lead, the physical basis Z_S^+ contains propagating modes with $J > 0$ and evanescent modes decaying toward $m \rightarrow +\infty$:

$$\Phi_S(N+1) = Z_S^+ t. \quad (53)$$

E. Stable finite-length propagation

The finite central bilayer must retain all eight transfer modes. Let V_+ and V_- denote the 8×4 matrices whose columns are the right-going/right-decaying and left-going/left-decaying central modes. The central boundary states are represented as

$$\Xi_L = V_+ a_+^L + V_- a_-^L, \quad (54)$$

$$\Xi_R = V_+ P_+(N) a_+^L + V_- a_-^R. \quad (55)$$

Here

$$P_+(N) = \text{diag}[(\lambda_1^+)^N, \dots, (\lambda_4^+)^N], \quad (56)$$

$$P_-(N) = \text{diag}[(\lambda_1^-)^{-N}, \dots, (\lambda_4^-)^{-N}]. \quad (57)$$

The modes are referenced at the boundary where their explicit propagation factor has modulus no greater than one. This representation is analytically equivalent to direct propagation through T_C^N , but avoids generating exponentially large factors from evanescent modes.

V. INTERFACE MAPS AND CONDUCTANCE

A. Local interface maps

At each atomically clean contact, the layer that continues into the monolayer lead is matched by continuity, while the layer that terminates imposes the absence of the fictitious neighbor outside the physical domain. In the modal basis this produces one local 6×6 linear system at each interface. Solving the left interface gives

$$a_+^L = t_{\text{in}} + R_F a_-^L, \quad (58)$$

and

$$r = r^{(0)} + T_{F \leftarrow C} a_-^L. \quad (59)$$

Here t_{in} is the central amplitude injected by the incident ferromagnetic electron, R_F reflects a returning central mode back into the bilayer, and $T_{F \leftarrow C}$ extracts a central mode into outgoing ferromagnetic lead amplitudes.

At the superconducting contact,

$$a_-^R = R_S^{(g)} a_+^R, \quad t = T_{S \leftarrow C}^{(g)} a_+^R. \quad (60)$$

The geometry index g enters only through the right-contact selectors, because the right lead continues from layer 1 in the $1 \rightarrow 1$ geometry and from layer 2 in the $1 \rightarrow 2$ geometry. The matrix $R_S^{(g)}$ contains both normal and Andreev blocks in central modal space,

$$R_S^{(g)} = \begin{pmatrix} R_S^{ee,(g)} & R_S^{eh,(g)} \\ R_S^{he,(g)} & R_S^{hh,(g)} \end{pmatrix}. \quad (61)$$

The off-diagonal blocks arise from the Nambu structure of the superconducting lead modes; no phenomenological Andreev phase is inserted by hand.

B. Round-trip matrix

A right-going central amplitude leaving the left boundary first propagates to the right boundary, reflects at the superconducting contact, propagates back, and reflects at the ferromagnetic contact. The corresponding round-trip matrix is

$$\mathbf{M}_{\text{rt}}^{(g)}(E, K, \sigma, N) = R_F P_-(N) R_S^{(g)} P_+(N). \quad (62)$$

The exact cavity equation is

$$\left[\mathbb{I}_4 - \mathbf{M}_{\text{rt}}^{(g)}(E, K, \sigma, N) \right] a_+^L = t_{\text{in}}. \quad (63)$$

All matrices in Eq. (62) are 4×4 , while t_{in} and a_+^L are four-component central modal vectors.

Solving Eq. (63) and inserting the result into Eq. (59) gives the Andreev-reflected amplitude

$$r_{he}^{(g)} = e_h^\dagger T_{F \leftarrow C} P_-(N) R_S^{(g)} P_+(N) \times \left[\mathbb{I}_4 - \mathbf{M}_{\text{rt}}^{(g)} \right]^{-1} t_{\text{in}}. \quad (64)$$

Here $e_h = (0, 1)^T$ selects the hole component of the outgoing ferromagnetic lead amplitude. The direct reflection term at the left contact has no hole component because the left lead is normal.

The quasiparticle transmission into the superconducting lead is

$$t^{(g)} = T_{S \leftarrow C} P_+(N) \left[\mathbb{I}_4 - \mathbf{M}_{\text{rt}}^{(g)} \right]^{-1} t_{\text{in}}. \quad (65)$$

For $|E| < \Delta_0$, the physical superconducting modes are evanescent and carry no asymptotic transmitted flux.

C. BTK conductance

With all propagating lead modes normalized to unit flux, the reflection and transmission probabilities are

$$R_N = |r_{ee}|^2, \quad R_A = |r_{he}|^2, \quad T_{\text{QP}} = \sum_{\nu \in \text{prop.}} |t_\nu|^2. \quad (66)$$

Flux conservation gives

$$R_N + R_A + T_{\text{QP}} = 1 \quad (67)$$

on the support of the incident channel. The single-channel BTK conductance is

$$g(E, K, \sigma, N) = \chi_{\text{in}}(E, K, \sigma) \left[1 - R_N(E, K, \sigma, N) + R_A(E, K, \sigma, N) \right]. \quad (68)$$

where

$$\chi_{\text{in}}(E, K, \sigma) = \begin{cases} 1, & \text{incoming electron channel open,} \\ 0, & \text{otherwise.} \end{cases} \quad (69)$$

The condition $\chi_{\text{in}} = 1$ means that a propagating electron eigenmode of T_F with $J > 0$ exists at the chosen (E, K, σ) . Equivalently, using Eq. (67),

$$g(E, K, \sigma, N) = \chi_{\text{in}}(E, K, \sigma) \left[2R_A(E, K, \sigma, N) + T_{\text{QP}}(E, K, \sigma, N) \right]. \quad (70)$$

The dimensionless conductance per transverse unit cell, in units of e^2/h , is. In Eq. (71), $R_A^{(g)}$ and $T_{\text{QP}}^{(g)}$ are evaluated at the same arguments (E, K, σ, N) as χ_{in} .

$$G^{(g)}(E, N) = \frac{1}{2\pi} \sum_{\sigma=\pm 1} \int_{-\pi}^{\pi} dK \chi_{\text{in}}(E, K, \sigma) \times \left[2R_A^{(g)} + T_{\text{QP}}^{(g)} \right]. \quad (71)$$

For subgap energies,

$$|E| < \Delta_0, \quad T_{\text{QP}}^{(g)} = 0, \quad (72)$$

and hence

$$G_A^{(g)}(E, N) = \frac{1}{2\pi} \sum_{\sigma=\pm 1} \int_{-\pi}^{\pi} dK \chi_{\text{in}}(E, K, \sigma) \times 2 \left| r_{he}^{(g)}(E, K, \sigma, N) \right|^2. \quad (73)$$

For a system of N_y transverse cells, the total conductance is $(e^2/h)N_y G^{(g)}$, and the conductance per physical width $W = N_y a_y$ is $(e^2/h)G^{(g)}/a_y$.

VI. RESONANCE INTERPRETATION

A. Real-axis resonance diagnostics

The exact object controlling internal resonances is the round-trip matrix in Eq. (62). Since the scattering amplitudes contain the resolvent $[\mathbb{I}_4 - \mathbf{M}_{\text{rt}}^{(g)}]^{-1}$, a resonance is associated with near-singularity of

$$\mathcal{R}^{(g)}(E, K, \sigma, N) = \mathbb{I}_4 - \mathbf{M}_{\text{rt}}^{(g)}(E, K, \sigma, N). \quad (74)$$

The determinant

$$D_g(E, K, \sigma, N) = \det \mathcal{R}^{(g)}(E, K, \sigma, N) \quad (75)$$

is useful analytically, but on the real energy axis the more robust numerical diagnostic is the smallest singular value

$$s_{\text{min}}^{(g)}(E, K, \sigma, N) = s_{\text{min}} \left[\mathcal{R}^{(g)}(E, K, \sigma, N) \right]. \quad (76)$$

A small value of s_{min} indicates that a pole of the open scattering problem lies close to the real axis. It does not by itself guarantee a visible peak in the conductance, because the corresponding pole may be weakly excited or weakly read out as a hole.

B. Round-trip eigenvalues and bright versus dark poles

Assume for the moment that $\mathbf{M}_{\text{rt}}$ is diagonalizable. Let

$$\begin{aligned} \mathbf{M}_{\text{rt}} v_j^R &= \mu_j v_j^R, \\ (v_j^L)^\dagger \mathbf{M}_{\text{rt}} &= \mu_j (v_j^L)^\dagger, \\ (v_j^L)^\dagger v_k^R &= \delta_{jk}. \end{aligned} \quad (77)$$

Then the resolvent gives the modal decomposition

$$r_{he}^{(g)} = \sum_j \frac{C_j^{(g)}}{1 - \mu_j^{(g)}}. \quad (78)$$

The residue factor is

$$C_j^{(g)} = \left[e_h^\dagger T_{F \leftarrow C} P_- R_S^{(g)} P_+ v_j^R \right] \left[(v_j^L)^\dagger t_{\text{in}} \right]. \quad (79)$$

Writing

$$\mu_j = \rho_j e^{i\Theta_j}, \quad (80)$$

ρ_j measures the survival of the amplitude after one complete internal cycle, while Θ_j is the coherent round-trip phase. The factor C_j is the product of an injection weight and a hole-readout weight. Thus a pole-like feature of $\mathbf{M}_{\text{rt}}$ is bright in $G(E, N)$ only if $|C_j|$ is appreciable. A mode for which $\mu_j \simeq 1$ but $C_j \simeq 0$ is a dark pole: it is present in the internal cavity but nearly invisible in the measured Andreev conductance.

For a single isolated bright mode, the approximate fixed-channel resonance condition is

$$\begin{aligned} \rho_j(E, K, \sigma, N) &\simeq 1, \\ \Theta_j(E, K, \sigma, N) &\simeq 2\pi\ell, \quad \ell \in \mathbb{Z}. \end{aligned} \quad (81)$$

This criterion refers to the channel-resolved quantity $g(E, K, \sigma, N)$, not yet to the transverse-integrated conductance.

C. From fixed-K resonances to integrated ridges

A ridge in $G^{(g)}(E, N)$ is a feature of the integrated quantity in Eq. (71). Therefore a fixed- K resonance survives transverse integration only if neighboring open channels resonate at nearly the same energy. Let $E_{\ell j}(K, N; \sigma)$ be the solution of

$$\Theta_j(E, K, \sigma, N) = 2\pi\ell \quad (82)$$

for a chosen branch. The open incident-channel window is

$$\mathcal{W}_{\text{in}}(E, \sigma) = \{K \in [-\pi, \pi] : \chi_{\text{in}}(E, K, \sigma) = 1\}. \quad (83)$$

A practical K -flatness diagnostic is

$$\delta E_{\ell j}(N; \sigma) = \max_{K \in \mathcal{W}_{\text{in}}} E_{\ell j}(K, N; \sigma) - \min_{K \in \mathcal{W}_{\text{in}}} E_{\ell j}(K, N; \sigma). \quad (84)$$

If

$$\delta E_{\ell j}(N; \sigma) \lesssim \Gamma_{\ell j}(N; \sigma), \quad (85)$$

where $\Gamma_{\ell j}$ is the numerical linewidth of the corresponding conductance peak, then neighboring transverse channels reinforce each other and the branch is expected to appear as a visible ridge in $G(E, N)$. If $\delta E_{\ell j} \gg \Gamma_{\ell j}$, the fixed- K resonances are dispersed over energy and are washed out by the transverse integral.

A residue-weighted diagnostic is obtained from

$$w_j(E, K, \sigma, N) = \frac{|C_j(E, K, \sigma, N)|^2}{|1 - \mu_j(E, K, \sigma, N)|^2}. \quad (86)$$

This avoids overemphasizing formally resonant branches that are dark because their injection or hole-readout residue is negligible.

D. Numerical workflow for ridge validation

The formal interpretation becomes a testable result only after the predicted resonance branches are compared with the numerical conductance maps. The code should output, for each g , E , K , σ , and N :

- (i) the open-channel mask χ_{in} ;
- (ii) R_N , R_A , T_{QP} , and the unitarity error $1 - R_N - R_A - T_{\text{QP}}$;
- (iii) the fixed-channel conductance $g(E, K, \sigma, N)$ and integrated conductance $G^{(g)}(E, N)$;
- (iv) the round-trip matrix $\mathbf{M}_{\text{rt}}^{(g)}$ and $s_{\text{min}}(\mathbb{I}_4 - \mathbf{M}_{\text{rt}}^{(g)})$;
- (v) the eigenvalues $\mu_j = \rho_j e^{i\Theta_j}$, with phases unwrapped across the grid;
- (vi) the residues C_j and weights w_j ;
- (vii) the branch curves $E_{\ell j}(K, N; \sigma)$ and the flatness measure $\delta E_{\ell j}(N; \sigma)$.

The central figure supporting the resonance interpretation should overlay the predicted curves $E_{\ell j}(N)$ on the computed $G_A(E, N)$ map, retaining only branches with $\rho_j \simeq 1$, appreciable $|C_j|$, and sufficient K -flatness. The round-trip eigenvector should then be used to diagnose whether a visible ridge is predominantly normal Fabry-Pérot-like, Andreev-like, or a mixed electron-hole interference mode.

VII. NUMERICAL RESULTS

This section is the next construction target. The manuscript should not yet claim a final physical conclusion before the ridge diagnostics of Sec. VI have been computed and compared with the conductance maps.

The first results subsection should present numerical validation: the mode classification, the convergence with respect to the transverse grid, and the unitarity error

$$\epsilon_{\text{unitarity}} = |1 - R_N - R_A - T_{\text{QP}}|. \quad (87)$$

This audit should be shown before any physical interpretation of the maps.

The second subsection should show the subgap Andreev conductance maps

$$G_A^{(1 \rightarrow 1)}(E, N), \quad G_A^{(1 \rightarrow 2)}(E, N), \quad (88)$$

for the parameter set used in the code validation. The comparison between the two geometries should focus on which ridges are common, which are shifted, and which are enhanced or suppressed by changing the contacted layer at the superconducting side.

The third subsection should implement the resonance overlay. For the dominant bright branches, plot the numerically obtained conductance map together with the predicted curves $E_{\ell j}(N)$ extracted from the phase condition of Eq. (81), filtered by the K -flatness diagnostic of Eq. (84) and the residue weight of Eq. (86).

The fourth subsection should analyze the dependence on the exchange field. Since h shifts the electron and hole longitudinal momenta in the ferromagnetic lead, it changes both the open-channel window χ_{in} and the phase matching in the bilayer cavity. The natural observable is the evolution of $G_A(E, N)$ and of the dominant round-trip eigenvalues with increasing spin polarization.

VIII. CONCLUSIONS

This section will be finalized after the numerical maps and ridge overlays are inserted. The intended conclusion is that an atomically resolved transfer-matrix treatment of a ferromagnet–bilayer–graphene–superconductor junction produces a compact round-trip description of Andreev transport. The finite bilayer acts as a coherent cavity whose resonant structure is controlled by the eigenvalues and residues of $M_{\text{rt}}^{(g)}$, while the observed conductance ridges additionally require sufficient transverse-channel coherence to survive the K integration.

The final version should emphasize three points: first, the interface transparency is not fitted but follows from the lattice connectivity and layer termination; second, the stable propagation basis avoids direct ill-conditioned powers of the central transfer matrix; third, the $1 \rightarrow 1$ and $1 \rightarrow 2$ geometries provide distinct microscopic routes for Andreev reflection because the superconducting contact samples different layer components of the bilayer modes.

Appendix A: BdG commutators and pairing convention

This appendix records the convention needed to keep the two spin-singlet BdG sectors in the same algebraic

form. In a complete one-particle basis, including spin, the pairing matrix is antisymmetric,

$$\Delta_{ij} = -\Delta_{ji}. \quad (A1)$$

For onsite s -wave pairing the spatial, layer, and sublattice labels coincide, while the two spin labels are opposite; the spatial part is symmetric, but the full spin-orbital matrix remains antisymmetric.

For the general quadratic pairing term

$$\hat{H}_\Delta = \frac{1}{2} \sum_{ij} \left(\Delta_{ij} \hat{c}_i^\dagger \hat{c}_j^\dagger + \Delta_{ij}^* \hat{c}_j \hat{c}_i \right), \quad (A2)$$

the creation part gives

$$\left[\frac{1}{2} \sum_{ij} \Delta_{ij} \hat{c}_i^\dagger \hat{c}_j^\dagger, v_l \hat{c}_l \right] = \frac{1}{2} \sum_i v_l (\Delta_{il} - \Delta_{li}) \hat{c}_i^\dagger \quad (A3)$$

$$= \sum_i \Delta_{il} v_l \hat{c}_i^\dagger, \quad (A4)$$

where Eq. (A1) was used. The conjugate pairing part gives a contribution proportional to

$$\sum_i \Delta_{il}^* u_i \hat{c}_l. \quad (A5)$$

Thus the index order in the hole equation is fixed by

$$(\Delta^\dagger)_{li} = \Delta_{il}^*. \quad (A6)$$

With the quasiparticle operator of Eq. (22), the s_σ factor in the hole component cancels the spin-dependent sign of the singlet commutator. After this cancellation the sector equations can be written as

$$Eu_k = \sum_l h_{kl} u_l + \sum_i \Delta_{ki} v_i, \quad (A7)$$

$$Ev_l = -\sum_i h_{il} v_i + \sum_i \Delta_{il}^* u_i. \quad (A8)$$

Equation (A8) is the component form of the BdG block matrix in Eq. (24).

Appendix B: Transfer matrices and bilayer roots

This appendix stores the algebra that is too detailed for the main text. For a normal monolayer with scalar energy parameter x , the staggered state obeys

$$L_M \phi_{m+1} = R_M \phi_m, \quad (B1)$$

with

$$L_M = \begin{pmatrix} x & \xi \\ \eta^* & 0 \end{pmatrix}, \quad R_M = \begin{pmatrix} 0 & -\eta \\ -\xi^* & -x \end{pmatrix}, \quad (B2)$$

$$T_M = L_M^{-1} R_M.$$

For the normal AB bilayer with $x_j = \epsilon - V_j$, the recurrence is

$$L_{\text{BL}} \Psi_{m+1} = R_{\text{BL}} \Psi_m, \quad T_{\text{BL}} = L_{\text{BL}}^{-1} R_{\text{BL}}, \quad (\text{B3})$$

where

$$L_{\text{BL}} = \begin{pmatrix} x_1 & \xi & 0 & 0 \\ \eta^* & 0 & 0 & 0 \\ 0 & 0 & x_2 & \xi \\ t_\perp & 0 & \eta^* & 0 \end{pmatrix}, \quad (\text{B4})$$

$$R_{\text{BL}} = \begin{pmatrix} 0 & -\eta & 0 & -t_\perp \\ -\xi^* & -x_1 & 0 & 0 \\ 0 & 0 & 0 & -\eta \\ 0 & 0 & -\xi^* & -x_2 \end{pmatrix}. \quad (\text{B5})$$

A useful implementation check is

$$\det T_{\text{BL}} = \left(\frac{\eta}{\eta^*} \right)^2 = e^{2iK}, \quad (\text{B6})$$

valid away from $\eta(K) = 0$.

Using the Bloch form $A_j(m) = U_{Aj} e^{im\vartheta}$, $B_j(m) = U_{Bj} e^{im\vartheta}$ and defining

$$f(\vartheta, K) = \eta(K) + \xi e^{i\vartheta}, \quad s = |f|^2, \quad (\text{B7})$$

the bilayer secular equation is

$$(x_1^2 - s)(x_2^2 - s) - t_\perp^2 x_1 x_2 = 0. \quad (\text{B8})$$

The two solutions are

$$s_\nu(\epsilon) = \frac{1}{2} \left[x_1^2 + x_2^2 + \nu \sqrt{(x_1^2 - x_2^2)^2 + 4t_\perp^2 x_1 x_2} \right], \quad (\text{B9})$$

$\nu = \pm 1.$

Writing $\vartheta = K/2 + Q$, one obtains

$$s = t^2 [3 + 2 \cos K + 4 \cos(K/2) \cos Q], \quad (\text{B10})$$

which leads to the roots quoted in Eq. (43).

Appendix C: Lattice current and pseudo-unitarity

For a hopping bond crossing the cut between cells m and $m+1$, the particle current carried by an electron amplitude is

$$J_{m \rightarrow m+1} = 2 \text{Im} [\xi A^*(m) B(m+1)]. \quad (\text{C1})$$

This sign follows from the continuity equation with the hopping convention of Eq. (7). For a BdG vector, the hole contribution enters with the opposite sign because a hole amplitude represents a missing electron. This gives the flux metrics in Eqs. (48) and (49).

If the current is conserved by one transfer step, then

$$(T\Phi)^\dagger \Sigma (T\Phi) = \Phi^\dagger \Sigma \Phi \quad (\text{C2})$$

for all states Φ , and hence

$$T^\dagger \Sigma T = \Sigma. \quad (\text{C3})$$

For transfer eigenvectors this implies

$$(\lambda_i^* \lambda_j - 1) \zeta_i^\dagger \Sigma \zeta_j = 0. \quad (\text{C4})$$

In particular, an evanescent mode with $|\lambda| \neq 1$ has zero self-flux, while distinct propagating modes can be chosen flux-orthogonal. Inside a degenerate propagating subspace, a stable numerical procedure is to diagonalize the Hermitian matrix $Z^\dagger \Sigma Z$ formed from any computed basis Z for the subspace, then normalize the resulting combinations to unit flux.

Appendix D: Interface matching systems

The local interface maps used in Sec. V are obtained from selector matrices that extract the active layer and the terminated-layer components from the central state Ξ . At the left contact, layer 1 continues into the ferromagnetic lead and layer 2 terminates. Let S_L extract the four layer-1 BdG components and M_L extract the two independent missing-neighbor components of the terminated layer. Then the homogeneous left-interface problem is

$$\begin{pmatrix} S_L V_+ & -Z_F^- \\ M_L V_+ & 0_{2 \times 2} \end{pmatrix} \begin{pmatrix} a_+^L \\ r \end{pmatrix} = - \begin{pmatrix} S_L V_- \\ M_L V_- \end{pmatrix} a_-^L. \quad (\text{D1})$$

Adding the physical incident state replaces the right-hand side of the active-layer equation by $\zeta_{\text{in}} - S_L V_- a_-^L$, leading to the affine map in Eq. (58).

At the right contact, the active selector depends on the geometry. For $g = 1 \rightarrow 1$, the superconducting lead continues from layer 1; for $g = 1 \rightarrow 2$, it continues from layer 2. The right-interface system is

$$\begin{pmatrix} S_R^{(g)} V_- & -Z_S^+ \\ M_R^{(g)} V_- & 0_{2 \times 2} \end{pmatrix} \begin{pmatrix} a_-^R \\ t \end{pmatrix} = - \begin{pmatrix} S_R^{(g)} V_+ \\ M_R^{(g)} V_+ \end{pmatrix} a_+^R. \quad (\text{D2})$$

The full 6×6 systems in Eqs. (D1) and (D2), solved with pivoted linear algebra, are the numerical definition of the contact maps. Schur-complement reductions are useful analytically, but a singular chosen pivot does not by itself imply that the full contact problem is singular.

- [2] G. E. Blonder, M. Tinkham, and M. T. Klapwijk, “Transition from metallic to tunneling regimes in superconducting microconstrictions,” *Phys. Rev. B* **25**, 4515 (1982).

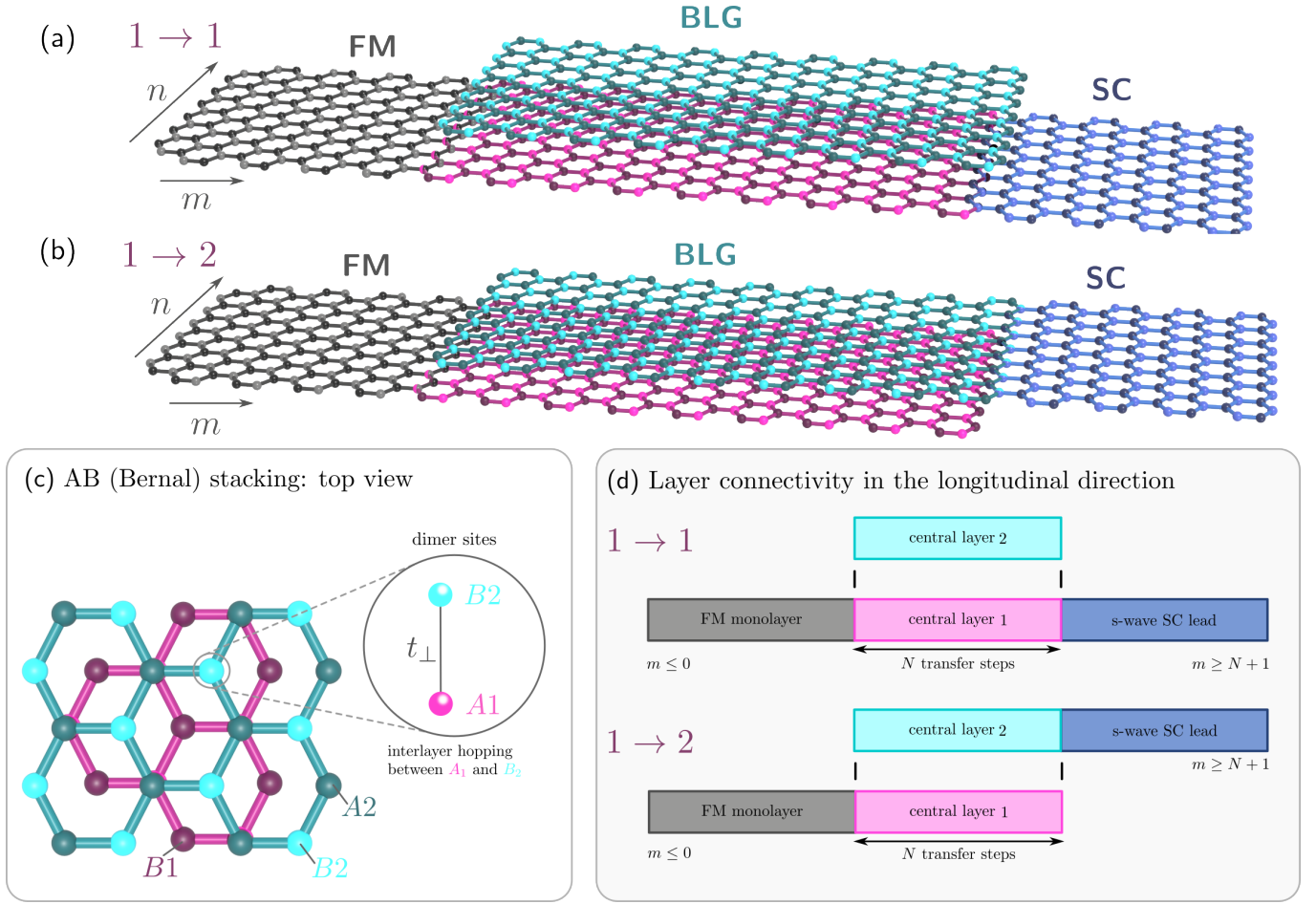


Figure 1. Geometry, lattice structure, and longitudinal connectivity of the ferromagnet–bilayer-graphene–superconductor (FM–BLG–SC) junction. (a) Atomic representation of the $1 \rightarrow 1$ geometry, in which the ferromagnetic (FM) and superconducting (SC) monolayer graphene leads are connected to the same layer of the finite bilayer graphene (BLG) region. (b) Corresponding $1 \rightarrow 2$ geometry, in which the FM and SC leads are connected to opposite graphene layers. The longitudinal and transverse lattice directions are denoted by m and n , respectively. Gray, magenta, cyan, and indigo identify the FM monolayer, central layer 1, central layer 2, and SC monolayer, respectively. (c) Top view of the AB (Bernal) stacking in the central bilayer. Different shades within each layer distinguish the A and B sublattices. The enlarged detail shows the vertically registered A_1 and B_2 dimer sites, coupled by the interlayer hopping $t_{\perp}$; the A_1 site is hidden beneath B_2 in the top projection. The non-dimer sites B_1 and A_2 are also indicated. (d) Longitudinal layer-connectivity representation of the $1 \rightarrow 1$ and $1 \rightarrow 2$ junctions. The FM lead occupies $m \leq 0$, the finite BLG region extends over N transfer steps, $1 \leq m \leq N$, and the SC lead occupies $m \geq N + 1$. In the $1 \rightarrow 1$ geometry the SC lead continues from central layer 1, whereas in the $1 \rightarrow 2$ geometry it continues from central layer 2.